LAB 1 – REQUIREMENTS ENGINEERING

Project: Urban Rooftop Gardening Co-op App
Problem Statement: #65

Functional Requirements

Req ID	Type	Description	Priority
FR-001	Functional	The system shall allow gardeners to log seasonal crop planting dates and calculate estimated harvest windows and expected yield weight.	High
FR-002	Functional	The system shall allow co-op gardeners to register rooftop plots by providing plot details, location and available growing area.	High
FR-003	Functional	The system shall allow gardeners to record the actual crop yield, including crop name, harvest date and harvested weight.	High
FR-004	Functional	The system shall allow gardeners and community leaders to create, view and share soil maintenance tips.	Medium
FR-005	Functional	The system shall allow gardeners to list surplus produce, view available listings and initiate produce trades with other community members.	High

Non- Functional Requirements

Req ID	Type	Description	Priority
NFR-001	Performance & Security	The system shall update available surplus produce listings in real-time across community members while protecting listing and user data.	High
NFR-002	Usability	The system shall provide an easy-to-use interface that allows gardeners to register a rooftop plot and record crop information without technical assistance.	Medium

Acceptance Criteria and Rationale

Req ID	Acceptance Criteria	Rationale
FR-001	Pass: Valid planting dates generate an estimated harvest window and expected yield. Fail: Invalid planting date sequences are accepted.	Helps gardeners plan crop harvesting.
FR-002	Pass: A valid plot registration is saved and assigned a unique plot ID. Fail: Duplicate plot IDs are generated.	Maintains organised rooftop plot records.
FR-003	Pass: Crop name, harvest date and weight are successfully stored and displayed. Fail: Invalid yield records are saved.	Tracks crop productivity.

Req ID	Acceptance Criteria	Rationale
FR-004	Pass: Users can publish and view soil maintenance tips. Fail: Published tips cannot be retrieved.	Encourages knowledge sharing.
FR-005	Pass: A surplus listing becomes visible to other community members after publishing. Fail: Published listings remain unavailable.	Encourages produce exchange and reduces waste.
NFR-001	Pass: Listing updates appear across connected users within 2 seconds under the defined test load, with unauthorised access blocked.	Ensures timely and secure trading.
NFR-002	Pass: At least 9 out of 10 test users complete plot registration without assistance.	Makes the application accessible to community gardeners.